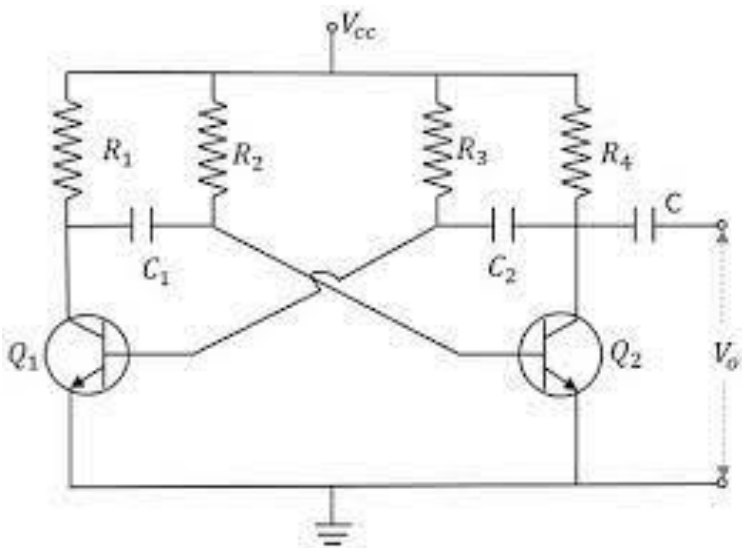


An astable multivibrator is an electronic circuit that continuously oscillates between its high and low states without the need for an external triggering signal. It is also known as a free-running multivibrator. The term "astable" indicates that the circuit is not stable in either state; it constantly switches between two states.

The basic astable multivibrator circuit consists of two cross-coupled amplifying devices, typically transistors, and some passive components like resistors and capacitors. The most common configuration uses two transistors, often configured as a pair of complementary astable multivibrator.



Here's a simplified explanation of how an astable multivibrator works:

1. **Transistors:** The circuit typically uses two transistors, often configured as Two NPN Or PNP transistor. Let's call them Q_1 and Q_2 .
2. **Coupling Capacitors:** Capacitors are used to couple the output of each transistor to the input of the other. These capacitors allow the feedback needed for oscillation.
3. **Resistors:** Resistors are used to bias the transistors and set the charging and discharging times for the capacitors.
4. **Operation:**
 - Initially, one of the transistors is in the ON state (saturation) while the other is in the OFF state (cut-off).
 - The ON transistor allows current to flow through its collector resistor and charges its associated coupling capacitor.

- As the capacitor charges, the voltage at the base of the other transistor increases until it reaches the threshold for turning that transistor ON.
- The newly turned ON transistor then pulls the voltage at the base of the first transistor low, turning it OFF.
- The process repeats, with the circuit continuously oscillating between the two states.

5. **Frequency of Oscillation:** The frequency of the oscillation is determined by the values of the resistors and capacitors in the circuit. By adjusting these components, you can control the frequency of the output waveform.

The output waveform is typically a square wave, and the duty cycle (ratio of time the waveform is high to the total period) can be adjusted by changing the component values. Astable multivibrators are commonly used in applications such as clock generators, pulse-width modulation (PWM) circuits, and tone generators in electronic projects and devices.